Reverse a Singly Linked List

Problem statement

Construct a dynamically allocated singly linked list from N numbers in input order. Print its values in reverse order.

Input format

First line: N. If N > 0, a second line contains exactly N integers.

Output format

If N = 0 print EMPTY. Otherwise print reversed numbers separated by single spaces, then newline.

Constraints

• 0 <= N <= 100

• -10000 <= node value <= 10000

Sample input and output

Sample input	Expected output
5 1 2 3 4 5	5 4 3 2 1

Important boundary conditions

Use actual linked-list nodes, not just an array. Empty list is valid.

Implementation expectations

O(N) time is expected. Free every allocated node before exiting.

Submission requirements

Submit exactly one .c file. Use stdin and stdout without interactive prompts. Compile with GCC in C11 mode. Avoid undefined behavior, release dynamically allocated memory, and print precisely the required output. Follow the stated input constraints.

Teacher resources (not part of student handout)

A reference oracle, five sample student submissions, assignment-specific rules.json, and 12 verified tests are supplied separately in the instructor test pack. Do not distribute the reference oracle or deliberately faulty variants as student solutions.